

Different response mechanisms of yield strength and ultimate tensile strength in pure copper considering size effect

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ABSTRACT

The ultimate tensile strength (UTS) and the yield strength (YS) are two key performances of copper wire, which vary synergistically under macroscopic conditions. However, they exhibit different dimensional sensitivity at small dimensions, which provides new ideas for designing materials. This study tested the UTS of copper wire with different dimensions after annealing. It was found that the UTS became weaker in a smaller dimension. Dislocations in the surface grains moved quickly to the sample surface of the microstructure. The density of dislocations in the grains decreased, which significantly reduced the strength at small dimensions. The YS arises with smaller dimensions because the growth of hard grains increased the difficulty in opening the slip system. The evolution of the Taylor factor (M) with dimensions explains the size effect in YS. The rapid increase in M value for small-size samples resulted in higher YS.

1. Introduction

Copper and copper alloys have attracted research attention in the aerospace and electronic communication industries owing to their superior electrical and thermal conductivity. In electronics and engineering, copper and copper alloys are the functional and structural core units of electrically and thermally conductive components. Downsized components require thinner and finer structural copper. However, the properties of these finer and thinner metals do not change in equal proportion as the metal dimensions become micron size. This phenomenon led us to investigate the properties of smaller dimensions [1–4].

To understand size dependence, researchers have focused attention on the material size scale and its effect on flow stress of metals. Several studies have shown that the ultimate tensile strength (UTS) of pure copper and copper-zinc alloys decreases as the sample dimension increases [5,6]. Studies on copper foils by Wang [7–9] have shown that in heterogeneous surface layers, such as oxide layers and metal plating, the UTS increases with increasing sample dimension. Several studies have also revealed that grain size influences the yield strength (YS) of copper, steel, and aluminum, indicating that YS increases significantly with

reduced sample dimensions [10–12]. Studies have found that the decrease in the YS is related to the ratio of the sample dimension to grain size; a ratio of 1 is the lowest strength point, and the strength increases as the ratio moves away from 1, whether it increases or decreases [13, 14].

Possible mechanisms responsible for these size effects include piling-up of dislocations against grain boundaries (GBs), thickness dependence of mean-free dislocation path, and source limitation in thin films. Discrete dislocation plasticity uses a continuum description of the elastic lattice and retains individual dislocations as carriers of the plastic deformation [15–17]. Many theories on size effects focus on the role of these dislocations [18–20]. Nicola [18–20] achieved agreement for thickness dependence of Cu film YS between the range of sample dimensions and grain sizes. Uchic [21,22] attributed the increase in strength of single-slip Ni micropillars to the decrease in defect propagation and storage with reduced sample dimension. Greer [23] found a similar trend in multi-slip compression tests of gold nanopillars. Cheng's studies [24,25] revealed that dislocations mainly vanish at the grain boundaries in thinner films or small grains. Under these conditions, dislocations may escape from the crystal at the nearest free surface

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before multiplying and interacting with other dislocations. Thus, dislocations can only be generated by activating a fixed source, which implies high stresses on the crystal for dislocation motion [26,27].

However, adopting dimension in tensile properties is yet to be established due to the lack of extensive research on UTS and YS performances. In this study, the variation patterns of UTS and YS at small dimensions were analyzed separately. In addition, a more accurate predictive model of strength was developed by highlighting and reviewing their similarities and differences. Furthermore, the quantitative relations between the microstructure and strength were studied, which provides an in-depth understanding of the strengthening mechanism of material and its microstructure.

2. Experiment and method

2.1. Sample preparation

Electrolytic pure copper of the same composition was cast in the form of rods and plates with a purity of 99.99 wt%. The rod sample was produced by Youke (Henan, China), with the diameter of 16 mm, and it was cold drawn to different diameter wires. A stepped annealing regime for samples with different dimensions was used to obtain uniform grain size, and the annealing regime is shown in Table 1. The plate sample was produced by Aobote group (Shandong, China) with an original thickness of 5 mm and the plate sample was cold-rolled to different thicknesses, and the annealing regime was as below. In this paper, dimension was used to represent the diameter of copper wire and the thickness of the foil (see Table 2).

2.2. Tensile tests

The annealed copper wires were cut into different specimens with a gauge length of 100.0 mm. The tensile tests were performed at room temperature on the wires using an Instron testing machine(34SC-05). Tensile test reference standard GB/T 10573-2020, according to the elongation, choose the strain rate of $3.3 \times 10^{-4} \text{ s}^{-1}$. The UTS was the stress value corresponding to the maximum force at fracture. On the stress-strain curve, make a straight line parallel to the elastic section and translate 0.2 to the right, at which point the stress is $R_{p0.2}$. For alloys without a significant yield plateau, $R_{p0.2}$ is commonly used to represent the yield stress. The experiment was repeated five times to take the average value.

An in situ tensile test was employed to observe changes in the surface slip lines during the stretching process. The in-situ equipment requires a flat observation surface to prepare foil samples for in situ experiments. The test regime was the same as that of the wire samples after performing the tensile test. The in situ tensile observation was performed to confirm the similarity in the strength change pattern of the wire samples.

Foil samples for the in situ tensile test were cut (Fig. 1). The in situ tensile specimens were mechanically ground and then etched with a 6 g $\text{Fe}(\text{NO}_3)_3$ solution in 50 ml alcohol for 40–50 s. In situ tensile tests were performed in a CS-3400 type SEM (Hitachi, Japan) using a loading stage (maximum load capacity: 1 kN) inside the SEM chamber. The specimen was strained at a rate of $1 \times 10^{-5} \text{ s}^{-1}$.

2.3. Microstructure characterizing

The encapsulated copper wire was heated in a C-containing inlay. The encapsulated samples were polished with SiC paper, and these samples were vibration polished with the silicon dioxide suspension. The grain orientation maps and Taylor factor maps were measured and analyzed using the electron backscatter diffraction (EBSD) technique integrated into a ZEISS 7900F scanning electron microscope (SEM, Germany).

For TEM observation, the 200- μm wire samples were annealed at 300°C and then pre-stretched at 1.5% and 15% to obtain a precise dislocation observation. Thin slices of 5 μm were cut from the internal and surface of the sample using a focus ion beam (FIB) system. The sample surface was plated with platinum before cutting, and the surface grains could be distinguished according to the position of the platinum layer. The FIB sampling method is shown in Fig. 2. The TEM observation was carried out on an FEI Tecnai F20 microscope with an operating voltage of 200 KV.

3. Results and discussion

3.1. The contrast trends in UTS and YS

The UTS and YS trends of the copper wires of different diameters and grain sizes were measured (Fig. 3). The UTS of annealed specimens is observed to slightly decline with dimension decrease from 750 μm to 200 μm in Fig. 3a. As the dimension decreases under 200 μm , the UTS for specimens with a grain size of around 5.3 μm , 15.2 μm and 30.1 μm goes down by 27.1 MPa, 39.4 MPa and 64.4 MPa, respectively. UTS decreases rapidly at the dimension of 200 μm , and this decreasing trend is more obvious when grain size increases. This study concludes that the UTS are dimension and grain size-dependent. In Fig. 3b the YS is observed to slightly increase with the dimension decrease from 750 μm to 100 μm , and increase sharply at 100 μm . The YS increases synchronously with different grain sizes, and at certain dimensions, the specimen with a smaller grain size has higher YS, corresponding to the Hall-Petch relationship. Fig. 3 is the corresponding strain-stress curves with different sample dimensions and grain sizes (see Fig. 4).

A factor (D/d) of the ratio of sample dimension (D) and grain size (d) was introduced to reflect the size effect mechanism. In Fig. 3c, the relationship between UTS and D/d states that UTS first rises and then remains stable as D/d increases. The stable value for D/d of different grain size specimens differs, and this indicates that D/d is not the only parameter impacting the size effect. In Fig. 3d the YS increases sharply at D/d of 10.

At a certain wire diameter, an empirical formula (the Hall-Petch relationship) showed an inverse relationship between the grain size and YS, implying that a decrease in grain size increases the YS. However, there was no significant correlation between the UTS and grain size owing to the fracture mode driven by the tangential stress, which is a fully plastic fracture [28]. The dislocation movement led to the opening of the slip system, where dislocations continuously and persistently slipped and proliferated, resulting in plastic deformation that eventually reduced the sample cross-section down to a single point.

Table 1

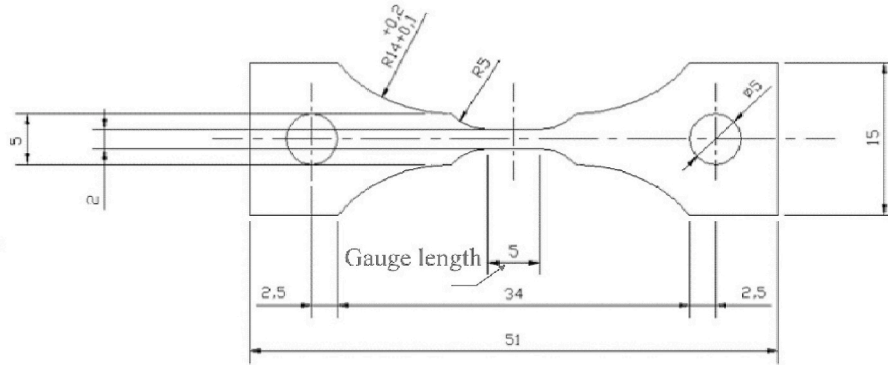
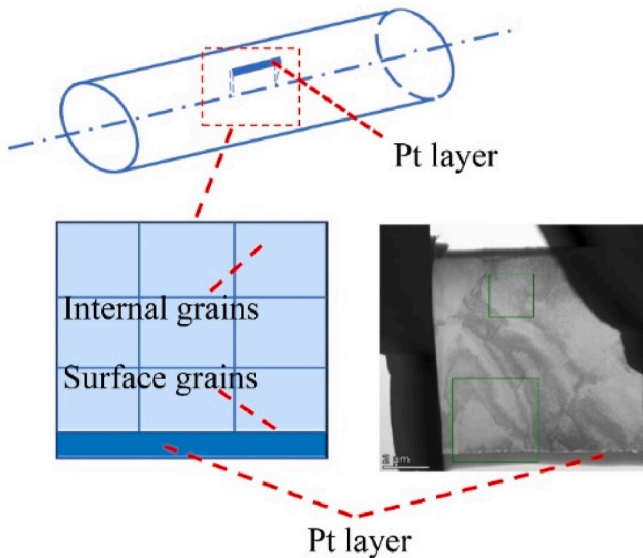
Grain sizes and corresponding annealing regimes for copper wires of different specimen dimensions.

Grain Size(μm)	Specimen Dimension(μm)							
	50	70	100	150	200	300	500	750
5.3–6.7	300°C×1 h	300°C×1 h	300°C×1 h	300°C×50 min	300°C×50 min	300°C×50 min	270°C×50 min	270°C×50 min
15.2–16.4	450°C×1 h	450°C×1 h	430°C×1 h	430°C×1 h	430°C×1 h	430°C×50 min	430°C×50 min	430°C×50 min
30.1–35.4	600°C×1 h	600°C×1 h	570°C×1 h	570°C×1 h	570°C×1 h	570°C×50 min	570°C×50 min	550°C×50 min

Table 2

Grain sizes and corresponding annealing regimes for film copper of different specimen dimensions.

Grain Size(μm)	Specimen Dimension(μm)						
	100	125	150	175	200	300	500
6.2–7.3	300°C×1 h	300°C×1 h	300°C×1 h	300°C×50 min	300°C×50 min	300°C×50 min	270°C×50 min
32.3–37.0	580°C×1 h	580°C×1 h	560°C×1 h	560°C×1 h	560°C×1 h	560°C×50 min	560°C×50 min

**Fig. 1.** Schematic image of in situ tensile specimens.**Fig. 2.** Schematic image of FIB cutting for transmission electron microscope (TEM) observation.

3.2. Size effect in UTS

According to the study, D/d affects the tensile strength because the surface grains differ from the internal grains by a dislocation reaction mechanism. Based on this, this study designed samples with grain size of 5.3–6.7 μm to obtain a clear grain organization with a low number of dislocations. After applying 1.5% pre-deformation, it was possible to characterize the phenomenon of dislocation appreciation and slip during dynamic tensile deformation. Fig. 5a shows the surface grains extracted by the FIB method, in which apparent linear dislocations and outcrops dislocations can be observed. The dark area around the punctiform defect is the stress distortion area [29], and it can also be observed along dislocation lines. Punctiform defects resulted in the deformation process with burger vector was perpendicular to the tensile

axis. Compared with the internal grains in Fig. 5b, the number of punctiform defects in the surface grains was significantly less. After contrast adjustment with Image Pro Plus (IPP) software, the black dislocation positions were highlighted. The dislocation maps of surface and internal grains are shown in Fig. 5c and d. The number of independent black areas was counted as individual dislocations. Line dislocations and point dislocations were all included in the statistics, and the number of dislocations per unit area (i.e., dislocation density) can be derived as shown in Fig. 5e. The dislocation density of surface and internal grains were $2.71 \times 10^{14} \text{ m}^{-2}$ and $3.95 \times 10^{14} \text{ m}^{-2}$, respectively. The varied numbers indicate that the degree of dislocations induced in grain surfaces was limited during the tensile deformation process, which may be attributed to the annihilation of dislocations when they move to the sample surface.

In order to observe the dislocation lines moving towards the sample surface, in situ stretching experiments were performed. Figs. 6–7 show the typical tensile-displacement curves and in situ scanning electron microscope (SEM) images taken at sequential load levels during room temperature deformation. Each SEM image corresponds to a point on the curve (Fig. 6). To collect SEM images, the tensile process and load were paused and held, respectively, causing stress relaxation slightly; it loads again after imaging. Fig. 6 also shows some similarities between the in-situ stress-strain copper foils with grain size of 6.2–7.3 μm and 33.3–37.0 μm . Thus, there was no significant yield point after the elastic section. The late linear segment sign was taken from the beginning of the plastic deformation stage, and four strain degrees were taken as the study points.

Fig. 7 shows the microscopic diagram of different strain stages after plastic deformation. When plastic deformation occurs, the dislocations start moving towards the sample surface. An obvious parallel dislocation steps (i.e., slip bands) can be observed on the sample surface as the dislocations reach the sample surface. Fig. 7a shows the beginning of the plastic stage, and it can be seen that there was no slip band observed as the strain stage reached 1.2% (Fig. 7b). The first slip zone appears in the views. Adjacent grains are shown in Fig. 7c and d, and most of the grains in view show parallel slip bands in different directions. No slip bands were found in the annealed samples (6.2–7.3 μm grain size) at the stress level of 0.4% and 1.2% (Fig. 8a and b). However, grain elongation was

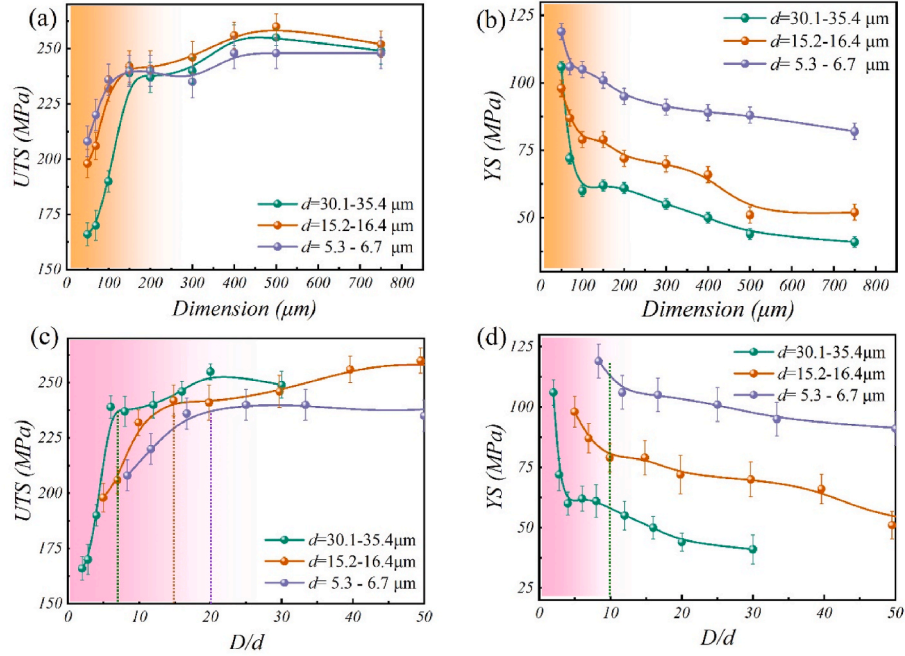


Fig. 3. The strength evolution of different sample grain sizes and sample dimensions. (a) UTS with dimensions; (b) UTS with D/d; (c) YS with dimensions; (d) YS with D/d.

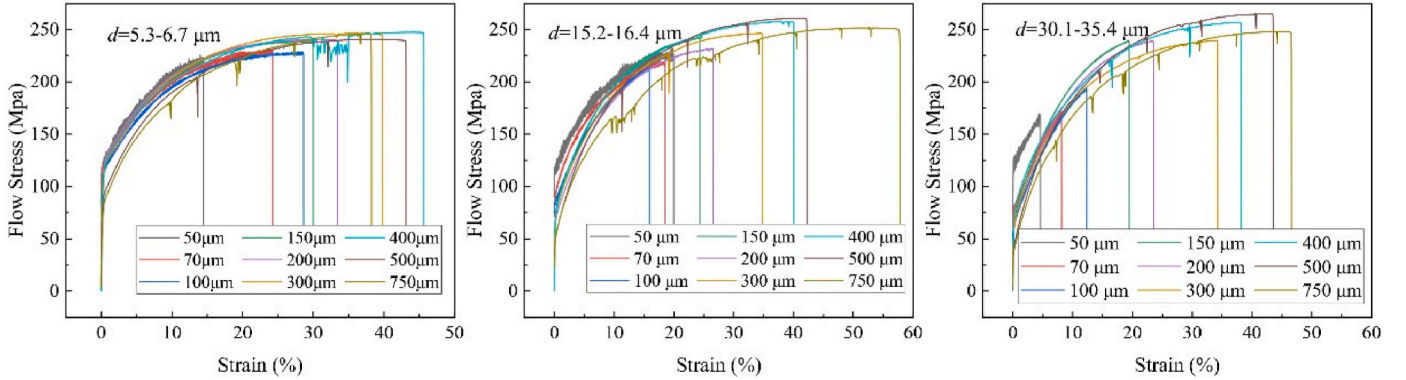


Fig. 4. Stress-strain curves of samples with different grain sizes and sample dimensions.

observed on the samples. Slip bands start to appear when the stress level reaches 2.0% (Fig. 8c). Few grains had slip bands when the stress level reached 3.3% (Fig. 8d). These indicate that dislocations moved quickly to the surface when the surface grains were more prominent, as grain size and sample dimension decrease, slip bands appear earlier.

Theories about the size effect in UTS can be classified into two types: the “smaller is stronger” and the “smaller is weaker” theories. The “smaller is stronger” type can be attributed to dislocation starvation in the material at nanometer scale. Since dislocations tend to escape from the sample surface, a lower density of dislocations resulted, which requires tremendous stress to crack the dislocation source [30–32]. The grains of the nano samples were tiny, and the length of the dislocations was of the same order of magnitude; thus, resulting in a lower degree of dislocations inside the grains [33]. Microminiature grains refer to grains with dimensions of a few nanometers to tens of nanometers, and dislocations are mismatches of tens to thousands of atoms that can reach lengths of a few nanometers to hundreds of nanometers, and when the grain is not large enough to accommodate a full dislocation, the

dislocation will quickly slip out of the grain. That’s why many research on nano-sized samples follow a “smaller is stronger” regulation.

In this study, the strength was shown to be weaker as the dimension decreases, that is, “smaller is weaker”. Since the polycrystalline samples were subject to the surface layer effects at the microscale. During the tensile process, dislocations generated at the grain boundaries were moved to the surface and then released from the surface. Moreover, statistical dislocation density was reduced due to a lack of entanglement between dislocations. The TEM results show the differences between the surface grains and internal grains in the number of dislocations. At this microscopic scale, the decrease in dimension or the increase in grain size can magnify the proportion of the surface grain area of the sample.

The following equation can model the flow stress in macroscale discoveries. This equation is an empirical formula, and the value of q is usually 1 to 1/2 [34,35].

$$\sigma(\epsilon) = \sigma_0 + k_d(\epsilon)d^{-q} \quad (1)$$

Where d is the grain size of the polycrystal, q is the hardening factor,

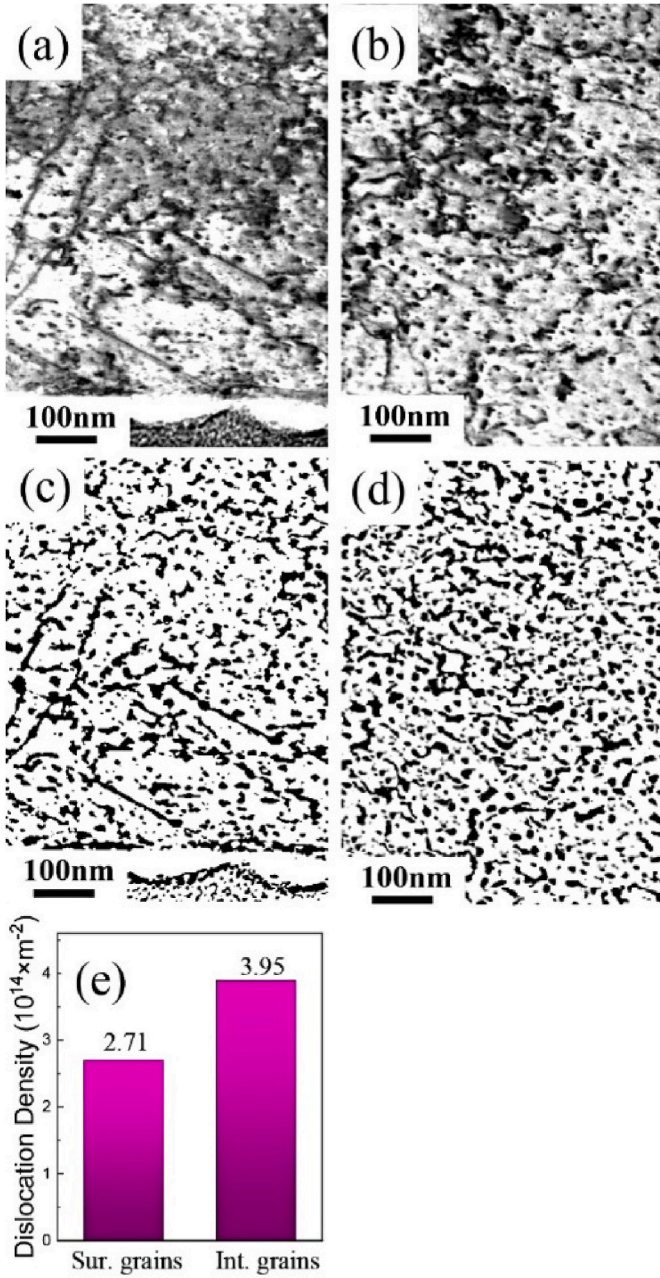


Fig. 5. TEM-STEM observation of annealed copper wire in a different position. (a) surface area; (b) internal area with a strain of 1.5%; (c) surface dislocations; (d) internal dislocations (e) counted number of dislocations.

σ_0 is the frictional force that prevents dislocation slip, and k is the strain-dependent factor. The coupling between the grain size and sample size dependence is complex. A scaling law of this type has been proposed.

$$\sigma(\varepsilon) = \sigma_0 + k_d(\varepsilon)d^{-q} + k_D(\varepsilon)D^{-p} \quad (2)$$

As Venkatraman and Bravman [34] proposed, the law does not provide a good fit for our result. Considering that the weakening effect of surface grains is related to the thickness grain ratio, and the equation can be changed to

$$\sigma(\varepsilon) = \sigma_0 + k_d(\varepsilon)d^{-q} + k_D(\varepsilon)(D/d)^{-p} \quad (3)$$

where D represents the sample dimension, and d represent the grain size. Fig. 9 demonstrates the fit of the experimental data achieved with these parameters. The simulated curves have almost coincided with the experiment curves, which remain stable at large dimensions and drop sharply at small dimensions. The stress of the material was affected by both grain size and sample dimension, which proves the feasibility of the equation model. The parameters used in the simulations are summarized in Table 3.

3.3. Size effect in YS

In strengthening and toughening theory, YS and UTS are generally considered together. These two strengths parameters are used to characterize the strength of a sample; however, in the plastic deformation stage, they are represented differently. The YS represents the beginning of homogeneous plastic deformation, while the UTS represents the end of homogeneous plastic deformation [36,37]. The YS needs to be discussed from the point of dislocating. The yield strain is the point at which yield strength occurs, and it is the starting point of uniform plastic deformation, i.e., the starting point of the slip system's opening and the starting point of initiating dislocation. At this point, the proposed quantitative model UTS is no longer applicable. However, the Taylor factor, which reflects the ease of opening the slip system, can be used to describe the yield strength change.

The Taylor factor is measured in the OIM system, with the slip system of $\{111\}\langle 110 \rangle$ and the critical stress (CRSS) of 0.980. Considering the performance of the single axial tensile test, the deformation gradient

$$\text{tensors } F \text{ is set as } \begin{bmatrix} 1 & 0 & 0 \\ 0 & -0.5 & 0 \\ 0 & 0 & -0.5 \end{bmatrix}.$$

The measure Taylor factor are shown in Table 4. The axial observation of the copper was performed to reveal the texture and the grain shape evolution. Fig. 10 shows the M values of different wire diameters. In 5.3–6.7 μm grain size curve, the value of the M increased from 3.0140 at 150 μm to 3.0424 at 50 μm with a decrease in the wire diameter, and this trend was consistent with YS. For 30.1–35.4 grain size specimens, the value of the M increased from 3.0213 to 3.1456, which indicated that the change in wire diameter affects the statistics of the M . These changes were pronounced in larger-grain samples. The change of M is synchronized with the change of $\langle 111 \rangle$ proportion.

The Taylor dislocation model gives the relationship between flow shear stress and dislocation density,

$$\tau = \alpha \mu b \sqrt{\rho} \quad (4)$$

where τ is the shear modulus, b is the magnitude of the Burger's vector, and α is an empirical coefficient, which is usually between 0.3 and 0.5.

Dislocation density is composed of two parts

$$\rho = \rho_s + \rho_G \quad (5)$$

where ρ_s represents the statistical storage dislocation density and ρ_G represents the geometrically necessary dislocation density. Huang [29] indicated that the geometrically necessary dislocation density is related to the curvature of plastic deformation or equivalent plastic strain gradient η^p

$$\rho_G = \bar{\tau} \frac{\eta^p}{b} \quad (6)$$

where $\bar{\tau}$ is the Nye factor. Arsenlis and Parks [38] introduced it to reflect the amplification factor of geometrically necessary dislocation density in three-dimensional nonuniform plastic deformation. For face-centered cubic crystals, $\bar{\tau}$ is approximately 1.90.

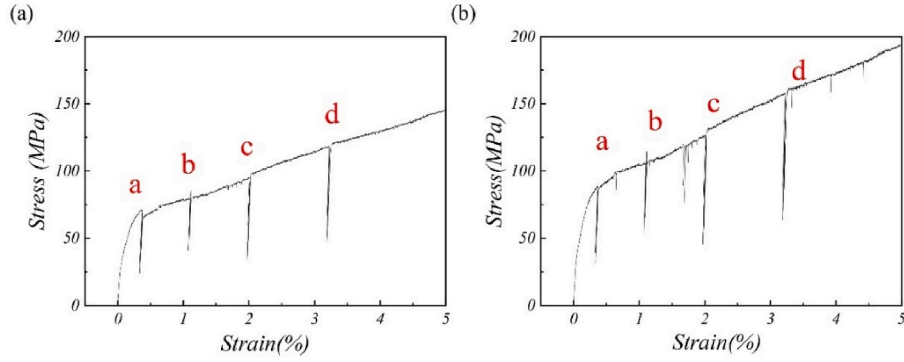


Fig. 6. In situ stress-strain curve of the copper foils with different grain sizes (a) 32.3–37.0 μm ; (b) 6.2–7.3 μm .

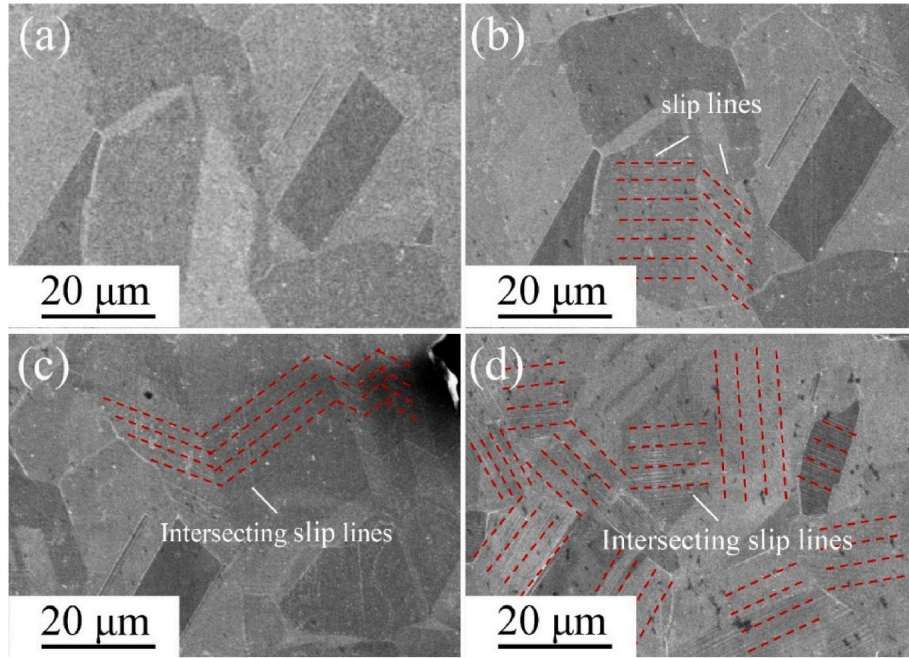


Fig. 7. SEM images of 32.3–37.0 μm copper foils during in-situ tensile test (a) 0.4%, (b) 1.2%, (c) 2.0%, (d) 3.3%.

The relationship between the tensile flow stress σ_{flow} and the shear flow stress τ is given as

$$\sigma_{\text{flow}} = M\tau \quad (7)$$

Substituting equations (4)–(6) into equation (7), we get

$$\sigma_{\text{flow}} = M\alpha\mu b \sqrt{\rho_s + \bar{r} \frac{l^p}{b}} \quad (8)$$

In the case of unidirectional stretching, the plastic strain gradient $\eta^p = 0$ if the flow stress at a single pull is $\sigma_{\text{ref}} f(\epsilon^p)$. From equation (6), we can determine the SSD density ρ_s , given as

$$\rho_s = \left[\frac{\sigma_{\text{ref}} f(\epsilon^p)}{M\alpha\mu b} \right]^2 \quad (9)$$

Bringing this into equation (5) gives the flow stress in the form of

$$\sigma_{\text{flow}} = \sqrt{[\sigma_{\text{ref}} f(\epsilon^p)]^2 + M^2 \bar{r} \alpha^2 \mu^2 b \eta^p} = \sigma_{\text{ref}} \sqrt{f^2(\epsilon^p) + l \eta^p} \quad (10)$$

Here

$$l = M^2 \bar{r} \alpha^2 \left(\frac{\mu}{\sigma_{\text{ref}}} \right)^2 b = 18 \alpha^2 \left(\frac{\mu}{\sigma_{\text{ref}}} \right)^2 b \quad (11)$$

where l is the endogenous material length in strain gradient plasticity ($M = 3.06$ and $\bar{r} = 1.9$ are used at the second equal sign above). For typical metallic materials, the burgers vector is on the order of a tenth of a nanometer, μ/σ_{ref} on the order of 10^2 , and α on the order of 0.3–0.5; thus, l should be on the order of micrometers, which is exactly the scale at which size effects appear in the experiment.

In this experiment, the size effect of the samples became more pronounced at the dimension of 100 μm , and the YS significantly increased at this dimension condition. This trend was exacerbated by the variation

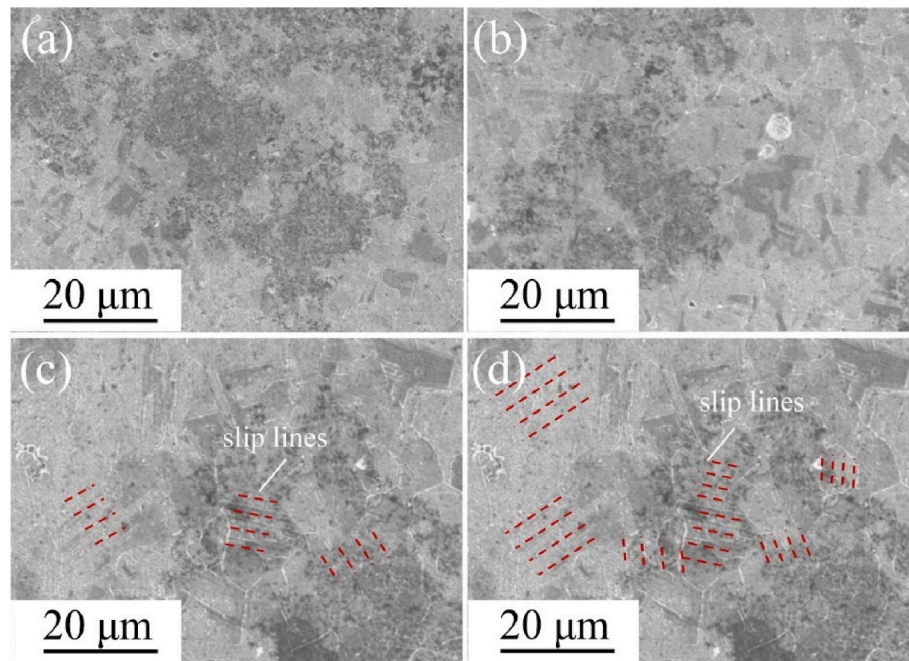


Fig. 8. SEM images of 6.2–7.3 μm copper foils during in-situ tensile test (a) 0.4%, (b) 1.2%, (c) 2.0%, (d) 3.3%.

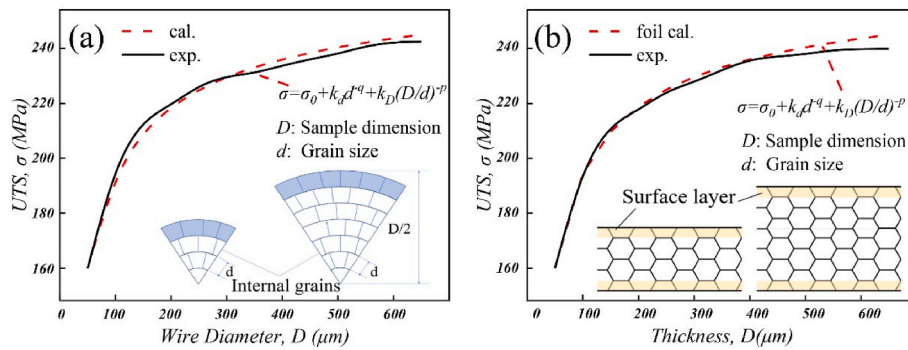


Fig. 9. Simulated graph of stress versus sample size (a) Wires with grain size of 5.3–6.7 μm ; (b) Foils with grain size of 6.2–7.3 μm .

Table 3
Values of the parameters used in the constitutive surface layer model.

	σ_0 (MPa)	k_d	k_D	q	p
Wire	60	1.68	1.73	0.54	0.87
Foil	57	2.03	1.61	0.54	0.79

Table 4
 M values and orientation information.

	50 μm	70 μm	100 μm	150 μm
M ($d=5.3\text{--}6.7\text{ }\mu\text{m}$)	3.0425	3.03697	3.03046	3.02875
M ($d=30.1\text{--}35.4\text{ }\mu\text{m}$)	3.14568	3.10481	3.03682	3.03217
<111> Proportion	0.51	0.44	0.42	0.22

of M values with wire diameter. The change of their weaving ratio led to the rises of the average Taylor factor (M) of the sample dimension. Moreso, large grain samples have larger grains, which are randomly oriented. The growth of these hard grains increased the overall statistical Taylor factor. Table 5 lists M values for macroscopic loading in the

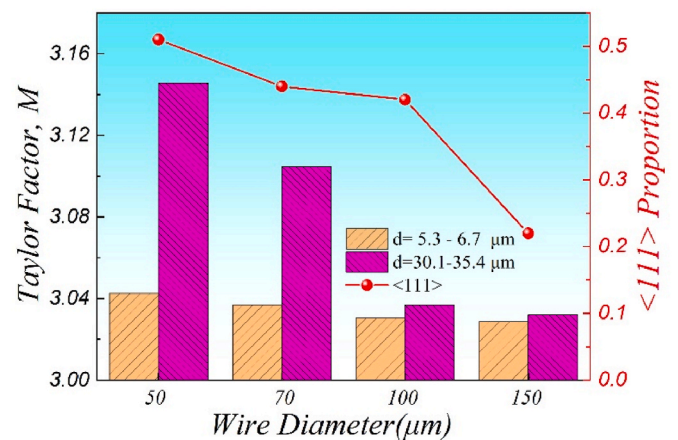


Fig. 10. Statistical chart of M evolution.

Table 5

M values for macroscopic loading in the drawing direction for various textures and activated slip systems [41].

Texture component	orientation	Taylor model	Sachs model	Hutchinson model [41]
Copper	{112}<111>	3.7	3.04	3.44
S	{123}<634>	3.33	2.5	2.97
Brass	{011}<211>	3.17	2.44	2.94
Cube	{001}<100>	2.45	2.45	2.45
Cube ND	{001}<110>	2.85	2.03	2.60
Random	/	3.07	2.24	2.9

drawing direction for various textures, which activated the slip systems. The value of *M* represents the minimum parting stress required for the slip system to open, which is a constant value related to the nature of the material itself. The stress is independent of the external force orientation [39,40].

4. Conclusion

In this study, different response mechanisms of YS and UTS in pure copper was investigated considering size effect. According to the experiment performed and mathematical modeling, we have conclusions as follow:

- (1) The UTS and YS result demonstrated a separate size effect with reduced dimensions (foils or wires) at micrometers. As sample dimension decreases, UTS increases and YS decreases.
- (2) The surface grains have a poor ability to accumulate dislocations from the microstructure. The movable dislocations escape from the sample surface, leading to a lower dislocation density, thus reducing the overall stress level of the material.
- (3) The change in YS is characterized by the variation in Taylor factor with the sample dimension. Slip system in samples with higher Taylor factor are difficult to be set into motion, bring about the rise in YS.

CRedit authorship contribution statement

Yu Hou: Investigation, Writing – original draft. **Wenjing Zhang:** Writing – review & editing. **Xujun Mi:** Funding acquisition, Supervision. **Haofeng Xie:** Funding acquisition, Supervision. **Xue Feng:** Data curation. **Guojie Huang:** Methodology. **Lijun Peng:** Methodology. **Zhen Yang:** Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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